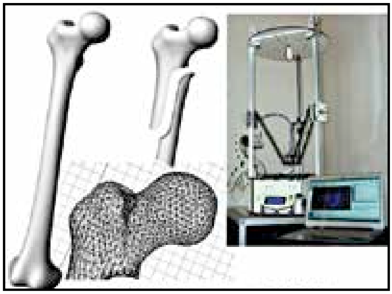
accurate mimic of the original bone piece.

The category of medical devices has also witnessed a revolution of its own with 3D printers having become the norm for most hearing aids that are being created. Since each hearing aid device is meant to fit an individual's ear, the ability to craft one based on scans from 3D printers makes this technology an obvious choice. Instead of having to spent time casting handmade ear moulds, a 3D scanner is able to save money to source the data required by the 3D printer. The time required is also reduced from over a week down to a single day. In the same way dental implants are also becoming faster, easier and more painfree to manufacture. One of the most revolutionary uses of 3D printing was in mid-2013 when doctors created a customised splint for a newborn infant who had suffered a collapsed trachea. The procedure relied on scans and 3D printing, which saved the child's life.



Knees and bone joint replacements are identical to the ones that are being replaced.

The third category of medical usage has been very controversial as it is involved with directly creating human tissues. The initial stages lead to the printing of meat tissue of animals that was considered suitable for eating and could be useful towards the world's hunger problems but the real victory has come with functional organic live human tissue.

Experiments to make a bionic ear using 3D printing are already underway and successful.

Scientists have used 3D printers to create functional liver tissue in the lab which are reactive for drug testing and surgical practice. Due to the ethical and moral concerns over the application of the technology, the move towards replacement organs is still slow but is predicted to be a reality some point in the future.